LESSON 10.7
SOLVING EQUATIONS INVOLVING THE DISTRIBUTIVE PROPERTY

LESSON INTRODUCTION

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.

LEARNING TARGETS

- I can solve two-step equations involving the distributive property.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

WORKING TOWARD

MA.8.AR.2.1 Solve multi-step linear equations in one variable with rational number coefficients. Include equations with variables on both sides.

ESSENTIAL QUESTIONS

- How do you apply the distributive property to solve two-step equations?
- How do you use the properties of equality to solve two-step equations?

LESSON NARRATIVE

In this lesson, students review the distributive property and apply it to solve two-step equations. Students learn that they can solve equations of the form $px+r=q$ either by using the distributive property or the division property of equality.

COMPONENT SUMMARY

10.7.1 WARM-UP *(~5 MIN.)*

Solve two-step equation puzzle

10.7.2 REFRESHER *(5 MIN.)*

Review distributive property

10.7.3 COLLABORATION *(5 MIN.)*

Justify why equations are equivalent based on properties of equality

10.7.4 GUIDED INSTRUCTION *(20 MIN.)*

Learn how to solve equations involving the distributive property

10.7.5 GUIDED PRACTICE *(5 MIN.)*

Practice solving equations involving the distributive property

10.7.6 COLLABORATION *(5 MIN.)*

Error analysis of solving equations involving the distributive property

10.7.7 WRAP UP *(~5 MIN.)*

Solving two-step equations

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Reciprocal
- Addition, subtraction, multiplication, and division Properties of equality
- Distributive property

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- Consider creating and displaying an anchor chart with the distributive property for students to reference.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize the equivalence of p(x+r) and (x+r)p.
- Students may forget to distribute to each term inside the parenthesis when using the distributive property.

THINGS TO CONSIDER

- Determine if your students need the support of all or part of the refresher.
- Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

10.7.4 Guided Instruction question 2 asks students to initially complete the problem independently and then discuss their work with their partners. The discussions will build student confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 10.7.3 Collaboration students engage in discussions that reflect on their own and their partners' mathematical thinking surrounding applications of the properties of equality and analyzing errors in samples of student work.

DIFFERENTIATE INSTRUCTION

In 10.7.5 Guided Practice, students use the distributive property to solve equations. To provide a visual entry-point, encourage students to draw arrows to each term inside the parenthesis.

10.7.1 WARM-UP: TWO-STEP EQUATIONS

TEACHER MOVES

Ask students to discuss what numbers they tried and the reasoning behind their attempts. Having students who successfully solved the puzzle share their strategies will help students that were unsuccessful. Emphasize that understanding how to solve the problem is more important than the solution.

10.7.1 Warm-Up: Two-Step Equations

1. Create an equation with the largest possible value for x by filling in the blue boxes using the digits 1 through 9. Each digit can only be used one time, at most. This may take several attempts. Show your work for each attempt you make.

$$\boxed{1}\,x - \boxed{9} = \boxed{8}$$

or

$$\boxed{1}\,x - \boxed{8} = \boxed{9}$$

10.7.2 Refresher: The Distributive Property

1. Several expressions are shown.

$(-2 \cdot x) + 3$ $\quad (x+3)(-2) \quad -2x - 6$

$(-2) \cdot x + (-2) \cdot 3 \quad (-2 + x)3$

a. Circle the expressions that are equivalent to $-2(x + 3)$.

b. Use substitution to show that each of the expressions chosen is equivalent to $-2(x + 3)$.

Let $x = 1$
$-2(1+3) = -8$
$(1+3)(-2) = -8$
$-2(1) - 6 = -8$
$(-2) \cdot 1 + (-2) \cdot 3 = -8$

2. Rewrite each expression by applying the distributive property.

a. $(-x + 6)4$
$-4x + 24$

b. $-3.5(-7 + 2y)$
$24.5 - 7y$

c. $\frac{7}{8}\left(16m - \frac{4}{5}\right)$
$14m - \frac{7}{10}$

10.7.2 REFRESHER: THE DISTRIBUTIVE PROPERTY

TEACHER MOVES

Recognizing the expressions are equivalent may be difficult for students. Help students see the structure in the distributive property by demonstrating how p(x+r) and (x+r)p are equivalent based on the commutative property of multiplication. Understanding equivalency of expressions and equations empowers students to think flexibly. Consider incorporating different equivalent expressions and equations as you work with students to model flexible thinking. Collaboration 10.7.3 supports student understanding of the equivalency of expressions.

DIFFERENTIATE INSTRUCTION

To provide a visual entry-point, encourage students to draw arrows to each term inside the parenthesis.

10.7.3 Collaboration: Equivalent Equations

1. All of the equations in the table below are equivalent. Work with your partner to explain how each equation is equivalent to the previous equation.

Equation	How is this equation equivalent to the equation above it?
$x = -6$	This is the given equation.
$x - 3 = -9$	*3 was subtracted from both sides of the equation.*
$-9 = x - 3$	*The symmetric property of equality was applied.*
$900 = -100(x - 3)$	*Both sides of the equation were multiplied by -100.*
$900 = -100x + 300$	*The distributive property was applied.*

2. Two students wrote equations equivalent to $2\left(m + 1\frac{1}{4}\right) = 3.5$.

Simon's Equation	Daphne's Equation
$2m + 2\frac{1}{2} = 3.5$	$m + 1\frac{1}{4} = 1.75$

a. Describe what Simon did to generate his equivalent equation.
Simon applied the distributive property and distributed the 2 to the m and the $1\frac{1}{4}$.

b. Describe what Daphne did to generate her equivalent equation.
Daphne divided both sides of the equation by 2.

10.7.3 COLLABORATION: EQUIVALENT EQUATIONS

DIFFERENTIATE INSTRUCTION

Allow students to reference an anchor chart with the properties of equality.

TEACHER MOVES

The students move from equivalency of expressions to equivalency of equations. Emphasize that the equations, $2(m+1\frac{1}{4}) = 3.5$, $2m + 2\frac{1}{2} = 3.5$, and $m+1\frac{1}{4} = 1.75$ are equivalent.

10.7.4 GUIDED INSTRUCTION: SOLVING EQUATIONS INVOLVING THE DISTRIBUTIVE PROPERTY

DIFFERENTIATE INSTRUCTION

If students write 65=72-p remind them to multiply 8 by both terms in the parenthesis. Students may feel more comfortable solving the equation if it is rewritten as 8(9-p)=65, because the variable is on the left side.

TEACHER MOVES

Students should be encouraged to utilize methods of solving equations that make sense for them. Although there are common methods that teachers consider to be easier, this may not be a method that a student considers easier. There are also students who do not like to be presented with options. Help these students embrace equivalency even if they would prefer not to consider alternate methods.

QUESTIONING

Consider challenging students who finish early to consider Partner C's work.

Partner C

$$8\frac{1}{2}=\frac{2}{3}(x+6)$$

$$\frac{8\frac{1}{2}}{\frac{2}{3}}=\frac{\frac{2}{3}(x+6)}{\frac{2}{3}}$$

TEACHER MOVES

Think-Pair-Share

Have students initially complete the question independently and then discuss their work with their partners. The discussions will build student confidence and understanding.

10.7.4 Guided Instruction: Solving Equations Involving the Distributive Property

Equations can be solved in more than one way. When an equation is written in the form $a(b+c)=d$, it can be solved by either distributing a first or by dividing both sides of the equation by a first.

1. The equation $65=8(9-p)$ can be solved using either method.

   a. Solve the equation $65=8(9-p)$ using each method.

Method 1 Apply the distributive property first	**Method 2** Divide both sides by the factor being distributed first
$65=8(9-p)$ $65=72-8p$ $65-72=72-72-8p$ $-7=-8p$ $\frac{7}{8}=p$	$65=8(9-p)$ $\frac{65}{8}=\frac{8(9-p)}{8}$ $8\frac{1}{8}=9-p$ $8\frac{1}{8}-9=9-9-p$ $-\frac{7}{8}=-p$ $-1*\left(-\frac{7}{8}\right)=-1*(-p)$ $\frac{7}{8}=p$

   b. Verify the solution is correct using substitution.

   $65=8(9-p)$
   $65=8\left(9-\left(\frac{7}{8}\right)\right)$
   $65=8\left(8\frac{1}{8}\right)$
   $65=65$

   c. Discuss each method of solving the equation with your partner. Then, explain which method you preferred for solving this equation.
   Answers will vary.

2. With your partner, decide who is Partner A and who is Partner B.

   a. Solve the equation $8\frac{1}{2}=\frac{2}{3}(x+6)$ using the method indicated.

Partner A Apply the distributive property first	**Partner B** Divide both sides by the factor being distributed first
$8\frac{1}{2}=\frac{2}{3}(x+6)$ $8\frac{1}{2}=\frac{2}{3}x+4$ $8\frac{1}{2}-4=\frac{2}{3}x+4-4$ $4\frac{1}{2}=\frac{2}{3}x$ $4\frac{1}{2}*\frac{3}{2}=\frac{3}{2}*\frac{2}{3}x$ $6\frac{3}{4}=x$	$8\frac{1}{2}=\frac{2}{3}(x+6)$ $\frac{3}{2}*8\frac{1}{2}=\frac{3}{2}*\frac{2}{3}(x+6)$ $12\frac{3}{4}=x+6$ $12\frac{3}{4}-6=x+6-6$ $6\frac{3}{4}=x$

   b. Share your solution and your work with your partner. Correct any errors, if necessary. Then, write your partner's method in the space provided once both partners agree the work is correct.
   Answers will vary.

   c. Explain which method you preferred for solving this equation.
   Answers will vary.

3. The perimeter of an equilateral triangle is 41.46 inches. If the length of each side is $(q - 5.7)$, the equation $3(q - 5.7) = 41.46$ can be used to find the value of q.

   a. What is the value of q?

   $3(q - 5.7) = 41.46$
   $3q - 17.1 = 41.46$
   $3q - 17.1 + 17.1 = 41.46 + 17.1$
   $3q = 58.56$
   $\frac{3q}{3} = \frac{58.56}{3}$
   $q = 19.52$

   b. The length of each side of the triangle is 13.82 inches because $41.46 \div 3 = 13.82$. Explain how the solution to the equation $3(q - 5.7) = 41.46$ can be used to show this as well.

   The length of one side is represented by $q - 5.7$, and q is equal to 19.52. The length of one side of the triangle is $19.52 - 5.7 = 13.82$. The side length can be multiplied by 3 to find the perimeter, 41.46.

10.7.4 GUIDED INSTRUCTION: SOLVING EQUATIONS INVOLVING THE DISTRIBUTIVE PROPERTY (CONTINUED)

QUESTIONING

Consider challenging students who finish early to solve the problems using both the distributive property and the division property of equality.

ENRICHMENT

- What are the features of an equilateral triangle?
- Which part of the equation tells us the length of one side of the triangle?
- How do you determine the perimeter of a triangle?

10.7.5 GUIDED PRACTICE: SOLVING EQUATIONS INVOLVING THE DISTRIBUTIVE PROPERTY

DIFFERENTIATE INSTRUCTION

To provide a visual entry-point, encourage students to draw arrows to each term inside the parenthesis.

10.7.5 Guided Practice: Solving Equations Involving the Distributive Property

Solve each equation using your preferred method. Then, verify that your solution is correct using substitution.

1. $\frac{3}{4}\left(k - 5\frac{1}{2}\right) = \frac{5}{8}$

$\frac{4}{3} * \frac{3}{4}\left(k - 5\frac{1}{2}\right) = \frac{5}{8} * \frac{4}{3}$

$k - 5\frac{1}{2} = \frac{5}{6}$

$k - 5\frac{1}{2} + 5\frac{1}{2} = \frac{5}{6} + 5\frac{1}{2}$

$k = 6\frac{1}{3}$

Check:

$\frac{3}{4}\left(6\frac{1}{3} - 5\frac{1}{2}\right) = \frac{5}{8}$

$\frac{3}{4}\left(\frac{5}{6}\right) = \frac{5}{8}$

$\frac{5}{8} = \frac{5}{8}$

2. $-3 = (w - 1.7)(-10)$

$-3 = -10w + 17$

$-3 - 17 = -10w + 17 - 17$

$-20 = -10w$

$\frac{-20}{-10} = \frac{-10w}{-10}$

$2 = w$

Check:

$-3 = -10(2) + 17$

$-3 = -20 + 17$

$-3 = -3$

10.7.6 Collaboration: Error Analysis

Work with your partner to complete the following.

1. Francesca was asked to solve the equation $-4\frac{1}{2} = \frac{1}{3}(x - 1.5)$. Her work is shown.

$$-4\frac{1}{2} = \frac{1}{3}(x - 1.5)$$
$$-4.5 + 1.5 = \frac{1}{3}(x - 1.5) + 1.5 \longleftarrow$$
$$-2 = \frac{1}{3}x \longleftarrow$$
$$\frac{-2}{\frac{1}{3}} = \frac{\frac{1}{3}x}{\frac{1}{3}}$$
$$x = -6$$

   a. Draw an arrow next to the step(s) where Francesca made a mistake.

   b. Describe the mistake(s) Francesca made.
   Francesca should have used either the distributive property or use the division property of equality first. Also, when -4.5 and 1.5 are added together, the sum is -3.

   c. Find the solution to $-4\frac{1}{2} = \frac{1}{3}(x - 1.5)$.
   $-4\frac{1}{2} = \frac{1}{3}(x - 1.5)$
   $3 * \left(-4\frac{1}{2}\right) = 3 * \frac{1}{3}(x - 1.5)$
   $-13\frac{1}{2} = x - 1\frac{1}{2}$
   $-13\frac{1}{2} + 1\frac{1}{2} = x - 1\frac{1}{2} + 1\frac{1}{2}$
   $-12 = x$

10.7.6 COLLABORATION: ERROR ANALYSIS

TEACHER MOVES

Highlight students' responses that refer to the distributive property, multiplication property of equality, and/or the division property of equality when explaining Francesca's mistake. Emphasize the importance of referring to the properties by name.

10.7.7 Wrap-Up: Ticket Out the Door

1. Solve $5\left(x + \frac{2}{5}\right) = {}^{-}13$.
   $5\left(x + \frac{2}{5}\right) = -13$
   $5x + 2 = -13$
   $5x + 2 - 2 = -13 - 2$
   $5x = -15$
   $x = -3$

10.7.7 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Consider tallying which methods your students choose to get a general idea of the overall preferred method in your class.